

Lead Generation System — Business Overview

How the system finds, qualifies, scores and prioritizes remote AI/ML hiring leads, and produces the final Excel deliverable.

Purpose

The system searches remote AI/ML job postings across the web, identifies the companies behind them, filters out companies unlikely to buy our services, and produces a ranked, contact-ready list of leads in a single Excel file.

1. How the Priority Score Is Calculated

Every qualified company is scored on a **0–100 scale** using six weighted factors (see `lead_signals.py`):

Factor	Max Points	What It Measures
ICP Fit (company size)	30	How closely the company's employee count matches our preferred customer size
Hiring Signal Strength	25	How strong the AI/ML hiring activity is (based on number of matching roles)
Contact Found	15	Whether a usable contact was found, and how confident we are in it
Multiple Openings	10	Whether the company has more than one matching job opening
Funding Signal	10	Whether the job posting itself mentions funding (e.g., "Series A")
New AI Initiative	10	Whether the company appears to be building/expanding an AI team, vs. already mature

Rules for each factor:

- **ICP Fit:** Full points for "Preferred ICP" size band; 60% for Medium; 25% for Large; 40% for Below-ICP; 50% if size is unknown (unknown is never treated as a bad sign).
- **Hiring Signal:** Full points for "Strong" signal, 60% for "Moderate," 20% for "Weak."
- **Contact Found:** 100% of points for High-confidence contact, 70% for Medium, 40% for Low, 0 if no contact found.
- **Multiple Openings:** Full points for 3+ matching openings, 60% for exactly 2, 0 for a single opening.
- **Funding Signal:** Full points if the job posting itself states funding language; 0 otherwise (no external funding research is performed).

- **New AI Initiative:** Full points if the company is "Building new AI capability," 60% if "Expanding AI team," 0 if it already has an "Established AI team," 30% if unclear.

All raw points are totaled and **normalized to a 0–100 score** ($100 \times \text{total points earned} / \text{total points possible}$). Every reason contributing to the score is recorded and stored for auditability.

2. How Leads Are Classified: High / Medium / Low

Based on the normalized 0–100 score:

- Score $\geq 70 \rightarrow$ **High Priority**
- Score $\geq 45 \rightarrow$ **Medium Priority**
- Score $< 45 \rightarrow$ **Low Priority**

Business-facing adjustment: To keep the delivered list action-oriented, the displayed priority tier is promoted one step before it reaches the final spreadsheet: High stays **High**, Medium becomes **High**, Low becomes **Medium**.

(The underlying numeric score is left unchanged in the export for transparency — only the labeled tier is promoted.)

3. Filters and Qualification Criteria

Before a company is counted as a lead, it passes through several checks:

a) Remote-only check

Every posting is re-verified as genuinely remote before it enters the pipeline.

b) Competitor exclusion

Companies are automatically excluded if their own name, industry, or description identifies them as one of the following (a direct competitor to our AI engineering/services offering), based on matched keyword phrases:

- Staffing / staff augmentation agencies
- Recruitment / executive search firms
- IT services / systems integrators
- AI/ML consulting firms
- Software outsourcing / development agencies
- BPO (business process outsourcing) providers

A company is also excluded if the job posting text itself indicates it was placed on behalf of a client (e.g., "our client is seeking..."), which marks the poster as a recruiting intermediary rather than the hiring company itself.

c) Company size banding

Companies are NOT excluded for size; instead they are banded and scored accordingly:

- **Preferred ICP:** 20–500 employees
- **Medium:** 500 employees up to the "low priority" threshold (5,000)
- **Large:** 5,000+ employees (kept, but scored lower)
- **Below ICP:** under 20 employees
- **Unknown:** kept and treated neutrally, never penalized as a bad fit

d) Persistent exclusion memory

Companies excluded in a previous run are remembered in a saved dictionary (`data/excluded_companies.json`) and automatically rejected in future runs without re-evaluating them.

e) Missing data policy

Any field the job boards did not supply (size, funding, contact, etc.) is reported as "Unknown" or "Not Found" rather than guessed or fabricated, and is never treated as disqualifying.

4. Data Sources / Websites Used

Job postings and company information are collected from the following platforms:

Source	Access Method
Indeed	Via JobSpy library
LinkedIn	Via JobSpy library
ZipRecruiter	Via JobSpy library
Glassdoor	Via JobSpy library
RemoteOK	Public JSON API (remoteok.com/api)
Remotive	Public JSON API (remotive.com/api/remote-jobs)
We Work Remotely	Public per-category RSS feeds
Jobspresso	Public RSS feed (jobspresso.co/jobs/feed)

Deliberately excluded (no legitimate public/free access exists): Wellfound/AngelList Talent (requires authenticated login; blocked by robots.txt) and FlexJobs (subscription paywall, terms prohibit automated collection).

Only postings from the **last 7 days (168 hours)** are considered by default.

5. APIs Used

Stage	API / Method
Job search & collection	JobSpy library (wraps Indeed, LinkedIn, ZipRecruiter, Glassdoor); RemoteOK public JSON API; Remotive public JSON API; We Work Remotely & Jobspresso RSS feeds
Company/industry enrichment	None external — all company size, industry, and description data comes directly from what the job boards already return with each posting
Contact extraction	No third-party contact-finder API is used. Contacts are extracted from: (1) emails already captured in the job posting text, (2) the company's own public careers page (fetched directly over HTTP, respecting robots.txt)
Contact "validation" / scoring	Rule-based scoring logic (not an external API) — ranks found emails by whether they sit on the company's own domain, contain recruiting-related wording (e.g., "careers@", "talent@"), or belong to an ATS vendor/compliance mailbox
LinkedIn contacts	Not scraped. The system generates LinkedIn people-search links (deep links a person can click), never an assumed or scraped profile

Note: There is no funding-data API, no email-verification API, and no third-party enrichment service integrated — every data point is sourced only from what the job boards and the company's own public web pages provide.

6. Overall Pipeline Flow

1. Job Collection	Search all configured platforms (Indeed, LinkedIn, ZipRecruiter, Glassdoor, RemoteOK, Remotive, WWR, Jobspresso) for the given job title and location, with similar-title keyword expansion.
2. Remote Filtering	Re-verify every posting is genuinely remote.
3. Company Identification & Deduplication	Group postings by company (via website domain, board profile URL, or normalized name) so each company appears once, with all its matching job openings combined.
4. ICP Filtering (Qualification)	Exclude direct competitors (staffing/recruiting/IT-services/AI-consulting/outsourcing/BPO firms) and previously-excluded companies. Band remaining companies by size (Preferred / Medium / Large / Below ICP / Unknown) — none of these are dropped.
5. Contact Enrichment	Extract contact emails from the job posting and the company's own careers page; generate LinkedIn search links; score and rank contacts by likely usefulness.
6. Lead Qualification (Signal Derivation)	Derive hiring-signal strength, AI-team maturity, funding mentions, and a plain-English lead summary for each company.
7. Lead Prioritization (Scoring)	Score every company 0–100 using the six weighted factors, assign High/Medium/Low priority, and rank all companies.
8. Final Selection & Excel Export	Cut the ranked list to the configured target number of leads (default: 50) and write the results to a single Excel workbook.

Final Excel Output — Tabs

- **Qualified Leads** — the final prioritized lead list
- **All Company Signals** — every scored company, for audit
- **Excluded Companies** — companies filtered out, with reasons
- **Pipeline Summary** — a stage-by-stage run report
- **Remote Jobs** — the underlying job-posting detail

Key Principles Behind the Design

- **No fabricated data.** If a data point (size, funding, contact) isn't available from a source, it is reported as "Unknown"/"Not Found" — never guessed or invented.
- **Full auditability.** Every score carries the specific reasons behind it, and every exclusion carries the matched phrase and source field that caused it.

- **Config-driven, not hard-coded.** All weights, size thresholds, and priority cut-offs live in `config.yaml`, so they can be retuned by a non-technical user without changing code.